

1. Real-World Devices Using SMPS and UPS

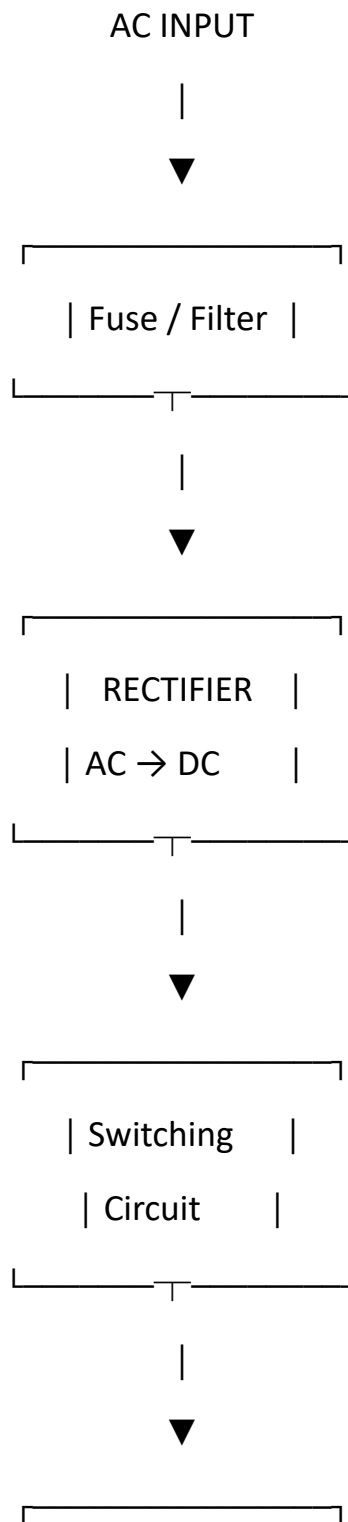
Devices using SMPS

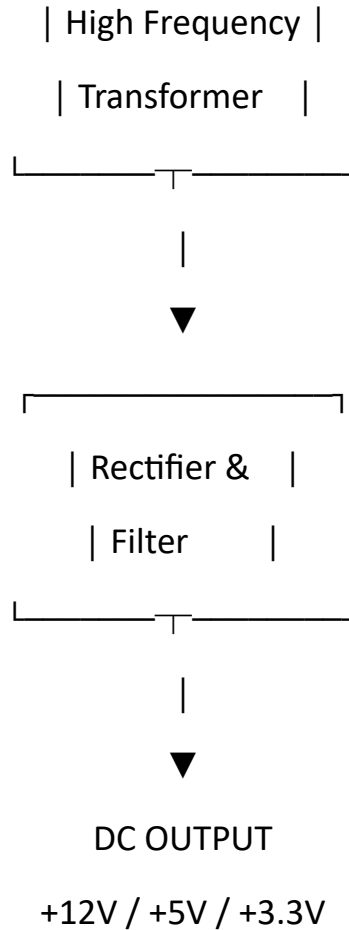
Device	Power Supply	Why it needs it
Desktop Computer	SMPS	Converts AC power into stable DC voltages required by computer components.
LED/LCD Monitor	SMPS	Provides regulated DC power to the display and internal circuits.
Wi-Fi Router	SMPS	Converts household AC power into low-voltage DC power for the router.

Devices using UPS

Device	Power Supply	Why it needs it
Desktop Computer	UPS	Provides backup power during power cuts and protects the PC from voltage fluctuations
Wi-Fi Router	UPS	Keeps the internet connection working for Some time during a power failure.

2. Block Diagram of a Typical SMPS





Main Components:

- AC Input: Receives electricity from the mains.
- Rectifier: Converts AC into DC.
- Switching Circuit: Switches the DC at high frequency.
- Transformer: Changes the voltage and provides isolation.
- Output Rectifier & Filter: Converts and smooths the output.
- DC Output: Supplies stable DC voltage to PC components.

3. Power Connectors

Connector	Powers	Safety Tip
ATX 24-pin	Motherboard	Turn off and unplug the PC before connecting or disconnecting it.
SATA Power Connector	HDD, SSD and other SATA drives	Do not force the connector; align it correctly before connecting.
Molex Connector	Older HDDs, optical drives and some accessories	Hold the connector firmly Hold the connector firmly and avoid pulling the cable.
PCIe 6/8-pin	Dedicated graphics cards (GPU)	Use the correct PCIe cable and make sure it is fully inserted.

4. Steps to Connect SMPS in a Gaming PC

Step 1: Install the SMPS

Place the SMPS inside the PC cabinet and secure it with screws.

Step 2: Connect the Motherboard

Connect the 24-pin ATX motherboard power connector from the SMPS to the motherboard.

Step 3: Connect the CPU

Connect the 4/8-pin CPU/EPS power connector to the CPU power socket near the processor.

Step 4: Connect the Graphics Card

If the graphics card requires additional power, connect the appropriate PCIe 6-pin, 8-pin, or 6+2-pin connector from the SMPS.

Step 5: Connect Storage Devices

Connect the SATA power connector from the SMPS to the HDD or SSD.

Step 6: Check All Connections

Make sure all connectors are properly inserted and cables are not interfering with fans.

Step 7: Power On

Connect the AC power cable to the SMPS, switch it on, and start the PC.

Important Safety Points

- Switch off and unplug the PC before changing power connections.
- Never force a connector into a socket.
- Use the correct cable for each component.
- Do not open the SMPS casing because it can contain dangerous electrical voltage.